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EDUCATION

The Pennsylvania State University <i>Ph.D. in Economics</i>	May 2025 <i>State College, PA</i>
Korea University <i>M.A. in Economics</i>	Feb 2019 <i>Seoul, South Korea</i>
Korea University <i>B.A. in Economics and Financial Engineering</i>	Feb 2017 <i>Seoul, South Korea</i>

ACADEMIC POSITIONS

Korea Development Institute <i>Associate Research Fellow</i>	July 2025 – Present <i>Sejong, South Korea</i>
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FIELDS OF INTEREST

International Trade, Industrial Organization

WORKING PAPER

Export Controls and Endogenous Upstream Price Incidence:
The Chip War's Consequences for the AI-Chip Industry

Modern semiconductor production is vertical, granular, and concentrated. This paper shows that in a bilateral oligopoly, export controls force a renegotiation of upstream input prices — a channel absent from standard quantitative trade models that treat input costs as fixed. I develop a vertical production network model in which AI-chip designers negotiate with a concentrated set of High-Bandwidth Memory (HBM) suppliers via Nash-in-Nash bargaining. Using Korean customs data, I document destination-specific price divergences consistent with relationship-specific bargaining. In the calibrated model, targeted buyers' input prices fall while rivals' input prices rise — the upstream shock-absorber and cross-buyer spillover channels. The input-price adjustment is roughly one-fifth of the downstream markup response, and a frozen-input counterfactual that ignores this renegotiation overestimates the competitive reallocation caused by export controls. Network composition matters: adding a single qualified supplier dampens the targeted buyer's markup increase by approximately 46 percent.

WORK IN PROGRESS

Trade, Multinational Production, and Regional Inequality
with Mira Rim

We study how China’s reductions in trade and multinational production (MP) barriers reshaped regional inequality in Korea. A shift-share IV design shows that omitting the MP channel attenuates the estimated import-competition effect; a multi-country, multi-sector GE model with trade, MP, and internal migration quantifies the 2000–2007 China shock, yielding an aggregate welfare gain of $\sim 2.0\%$ with per-worker gains of 1.7–2.5% across regions. A closed-form decomposition attributes 1.42 pp to trade, 0.55 pp to migration, and 0.04 pp to vertical MP. Jointly analyzing trade and MP is necessary both for causal identification and for accurate welfare accounting.

Triadic Gravity in Intermediated Petrochemical Trade

with Yang Shen

When firms can reroute shipments through intermediary hubs, the bilateral trade elasticity is strictly attenuated relative to the standard gravity benchmark — under Pareto productivity, by a factor equal to the share of trade flowing directly between origin and destination. Using transaction-level data from a Korean petrochemical trading house with 25 worldwide offices, we recover the structural attenuation parameter via a *Switch* control that absorbs mode selection. Standard gravity overstates route-specific trade-flow responses by $1.2\text{--}1.4\times$ at cross-sector-average intermediary shares and $1.7\text{--}2.4\times$ at shares plausible for petrochemical trade.

Equilibrium Uniqueness in Entry Games with Combinatorial Actions and Endogenous Pricing

with Sungwan Hong

We develop a uniqueness framework for combinatorial entry games with endogenous Atkeson–Burstein markups, decomposing the threshold-map Jacobian into density dampening, marginal-value sensitivity, and threshold location. When the resulting bathtub-shaped contraction modulus peaks below one, the Bayesian Nash equilibrium is unique and supermodularity is preserved so existing CDCP solvers remain applicable. In a quantitative application to global cloud infrastructure entry (five providers, 42 countries), the calibrated model matches 99.5% of in-sample entry decisions; endogenous markups sustain 15 additional firm-market pairs and lower consumer surplus by 10.6% relative to a constant-markup benchmark.

PUBLICATIONS

Kim, Hyungjin, and Jinhyuk Lee. “Collection of personal data with information externality in two-sided markets.” *Journal of Economic Theory and Econometrics* 31.3 (2020): 1–22. (*pre-PhD*)

RESEARCH EXPERIENCE

Research Assistant for Nonna Sorokina <i>The Pennsylvania State University</i>	2024–2025
Research Assistant for Jonathan Eaton <i>The Pennsylvania State University</i>	2022
Research Assistant for Seunghan Yoo <i>Korea University</i>	2017–2019

TEACHING EXPERIENCE

Instructor of Record

Summer 2021

The Pennsylvania State University

- ECON 302 Intermediate Microeconomic Analysis

Teaching Assistant

2020–2024

The Pennsylvania State University

- ECON 470 International Trade and Finance (SP24)
- ECON 434 International Finance (FA20, SP21, FA21, FA22, SP23)
- ECON 437 Multinationals and the Globalization of Production (SP22)
- ECON 333 International Trade (SU20)
- ECON 444 Economics of Corporation (SP20)

INVITED SEMINARS**2026** University of Pittsburgh**2025** Korea Development Institute; Bank of Korea; Korea University; Hanyang University; Korea Institute of Finance; Korea Insurance Research Institute; Korea Capital Market Institute; Korea Institute for International Economic Policy; Korea Institute for Industrial Economics and Trade; Korea Information Society Development Institute; Samsung Economic Research Institute; Center for International Development

CONFERENCE PRESENTATIONS**2026** Midwest International Trade Conference (Ohio State); IMF Economic Review (Bangkok, *scheduled*); Sapporo Workshop on Economic Theory (*scheduled*); Korea Academic Society of Industrial Organization (Seoul)**2025** Econometric Society World Congress (Seoul)**2024** American Economic Association annual meeting (San Antonio); Economics Graduate Student Conference (Washington University in St. Louis); Comparative Analysis of Enterprise Data (Penn State)**2023** International Economic Association World Congress (Medellin); Midwest International Trade (Georgia Tech); International Trade and Finance Association (Richmond); International Atlantic Economic Society (Philadelphia); CIREQ Ph.D. Conference (Université de Montréal)

AWARDS & HONORS**Fellow Paper, 33rd Annual Conference***International Trade and Finance Association*

2023

Doctoral Study Abroad Scholarship*Yongwoon Foundation*

2019

Brain Korea 21 Plus Scholarship*Korea University*

2017–2019

SPECIALIZED SKILLS**Programming Languages:** MATLAB, STATA**Language:** English (fluent), Korean (native), Japanese (beginner)

NATIONALITY

Citizenship: South Korea

REFERENCES

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